

# MATH3881/5881: Statistical Machine Learning Theory

## Week 3: Theory Tutorial: Dropout as Ensemble Averaging

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A single short problem to set up the Practice Tutorial. It draws on the last section of the Week 2 lecture notes (*Regularisation*, specifically the dropout subsection) and is the mathematical content behind Problem 2 of the Week 3 Practice Tutorial. Aim to finish it within the first 10 minutes of the tutorial so you have plenty of time for the notebook. A separate solutions document is provided a week later.

### Problem: Dropout is unbiased gradient descent in expectation

Let  $C : \mathbb{R}^d \rightarrow \mathbb{R}$  be a smooth loss function and  $\theta \in \mathbb{R}^d$  the current parameter vector. Recall from Week 2, §6, that *inverted dropout* with keep-probability  $q \in (0, 1]$  samples an i.i.d. Bernoulli mask  $m \in \{0, 1\}^d$  with  $m_i \sim \text{Bern}(q)$  and forms the dropout-modified gradient

$$\tilde{g} = \frac{1}{q} m \odot \nabla C(\theta),$$

where  $\odot$  is the elementwise (Hadamard) product. One dropout step is then  $\theta \mapsto \theta - \alpha \tilde{g}$ . The mask is independent of  $\theta$  and is freshly sampled at every step.

- (a) Compute  $\mathbb{E}[\tilde{g}]$ , where the expectation is over the mask only. Conclude that one inverted-dropout step is, *in expectation*, one standard gradient-descent step from Week 1 with step size  $\alpha$ .
- (b) Now consider the same scheme *without* the  $1/q$  rescaling, that is,  $\tilde{g}' = m \odot \nabla C(\theta)$ . Show that  $\mathbb{E}[\tilde{g}'] = q \nabla C(\theta)$ . Conclude in one sentence why PyTorch's `nn.Dropout` uses the  $1/q$  factor and not the raw mask.
- (c) *Connection to the Practice Tutorial.* Consider a *linear* model  $f(\theta; x) = \theta^\top x$ . Show that

$$f(\theta; x) = \mathbb{E}_m \left[ f \left( \frac{1}{q} m \odot \theta; x \right) \right],$$

exactly, where the expectation is over the dropout mask  $m$ .